MIL-G-50546B(AR)
21 December 1981
SUPERSEDING
MIL-G-50546A(AR)
10 March 1977

MILITARY SPECIFICATION

GRENADE, GP: M42
BODY ASSEMBLY, TYPE PH, FOR

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly, and packing for one type of body assembly designated as Body Assembly, Type PH, for use with Grenade, GP: M42.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

TT-C-490	-Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
MIL-H-6875	-Heat Treatment of Steels (Aircraft Practice, Process For)
MIL-I-45607	-Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS (see 6.7)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9215343	-Body Assembly Type PH
9206166	-Box, Fiber, Outer, for Body Assembly
9206167	-Box, Fiber, Inner, for Body Assembly

INSPECTION EQUIPMENT DRAWINGS

9266381	-Ultrasonic Test

2.2 Other Publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E-18	-Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.- The parts shall comply with all requirements specified on drawing (Dwg. 9215343), all associated drawings, and with all requirements specified in applicable specifications.

3.3 Identification and segregation of body assemblies - All body assemblies shall be recognizable as to homogeneity in accordance with the provisions of 4.4.1.

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3.4 Application of protective finish.- The specified protective finish shall not be applied until after the examination for cracks and voids is completed.

3.5 Records of results of contractor conducted examinations and tests. The contractor shall maintain records of all his examinations or tests required by this specification. (see 6.5)

3.6 First article inspection.- This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship.- All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for minor scratches not exposing base material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.- Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.- The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.- The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part description	drawings	quantity
Studs	9215345	80
Bodies (without protective finish)	9215344	80
Body Assemblies	9215343	80
Stock Specimens	9215344	2

4.3.2 Inspections to be performed.- See MIL-A-48078 and Table I herein.

4.3.3 Rejection

4.3.3.1 Attributes inspection.- If any component or assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3.3.2 Variables inspection-longitudinal and transverse force tests.- The first article sample shall be rejected if the results of the bodies tested in the Longitudinal Force Test or Transverse Force Test of Table I show either of the following:

a. Any body fails to comply with the minimum strength requirement specified on the applicable drawing, or

b. The value of the quantity ($\bar{X}$ - 3.0S), (i.e., the product of 3.0 and the sample standard deviation(S) of paragraph 6.6 subtracted from the sample average ($\bar{X}$)), is less than the applicable requirement.

NOTE: The value ($\bar{X}$ - 3.0S) shall be calculated once using the forty (40) longitudinal force test results and once using the forty (40) transverse force test results to determine compliance or noncompliance with the values specified on the drawing.

4.3.3.3 Ultrasonic inspection for cracks and voids.- If, during the first article test, any bodies are rejected by the automatic ultrasonic test system, such bodies shall be rerun thru the system again. If the previously rejected bodies are accepted, they shall be considered good and acceptable. If, however, a body is rejected a second time, that body shall be manually ultrasonically inspected for cause of rejection. The area of rejection shall be marked on the part. The rejected body shall then be sectioned, and the presence of a defect (i.e. crack or void) shall be verified. If a defect in any rejected body is verified, the first article sample shall have failed the ultrasonic inspection.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
	Grenade, GP: M42, Body Assembly, Type PH		SHEET OF		
	Studs (Dwg. 9215345) Examination for Defects	80	-	3.2	4.4.2.1
	Bodies (prior to forming or blanking Dwg. 9215344) Examination for Defects	2	-	3.2	4.4.2.2
	Bodies (without protective finish) (Dwg. 9215344) Examination for Defects	80	-	3.2	4.4.2.3
	Profile of Groove and Location from Open End of Body	80	-	3.2	4.5.2
	Hardness Test	*	-	3.2	4.5.3
	Longitudinal Force Test	40	-	3.2	4.5.4
	Transverse Force Test	40	-	3.2	4.5.5
	Body Assemblies (Dwg. 9215343) Examination for Defects	80	-	3.2	4.4.2.4

* 80 bodies shall be subjected to this test for informational purposes only.

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation.- Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of grenade assemblies shall contain:

a. Steel from not more than one manufacturer and of not more than one grade.

4.4.1.1 Heat treatment control.- When testing longitudinal and transverse strength in accordance with para. 4.4.3.3.a (continuous sampling) no specific controls will be assigned to the mill heat of material except by verifying that the material from any mill heat conforms to all the requirements in order to establish uniformity of the incoming material. When testing in accordance para. 4.4.3.3.b (lot sampling) a heat treatment lot shall consist of bodies from the same mill heat of material.

4.4.1.1.1 Bodies shall be heat treated by the same overall process (including the normalizing and/or stress relief phases, when part of the contractor's overall heat treat process) using the same quench bath and in the same furnace for each of the phases of the overall heat treatment process. Parts from each furnace shall be kept segregated until the tests specified in paragraphs 4.5.3, 4.5.4 and 4.5.5, of this specification have been completed. Beyond this point in the process, acceptable parts may be combined for further processing and lot formation after final operation in the process.

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4.4.2 Examination.- See MIL-A-48078.

a. Sampling plans.- Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-1235 plans CSP-1, CSP-2, CSP-T or MIL-STD-105, Inspection Level II.

b. Automatic gaging.- Facilities equipped with government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with contract.

c. Alternate inspection procedures.- The contractor may propose the use of mandatory alternate inspection procedures in lieu of conducting sampling inspection for defects listed in this specification. (See para 6.11).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-50546B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Stud		9215345
			NEXT HIGHER ASSEMBLY 9215343

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Inside Diameter, Maximum		.40%	3.2	Gage
102	Depth of Inside Diameter		.40%	3.2	Gage
103	Concentricity of Inside Diameter		.40%	3.2	Gage
104	Larger Outside Diameter, Minimum		.40%	3.2	Gage
MINOR					
201	Chamfer on Inside Diameter Improper		.65%	3.2	Visual
202	Evidence of Poor Workmanship		.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-50546B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Bodies (prior to forming or blanking)		9215344
			NEXT HIGHER ASSEMBLY
			9215343

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined				
MAJOR					
101	Dimensional inspection of stock specimen	*		3.2	4.5.1
MINOR	None defined				

NOTES:
* See applicable test method paragraph.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-50546B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.3	Body (prior to application of protective finish)		9215344
			NEXT HIGHER ASSEMBLY 9215343

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Inspection for cracks and voids		100%	3.2	4.5.5
MAJOR					
101	Largest outside diameter, min.		.40%	3.2	Gage
102	Largest inside diameter, max.		.40%	3.2	Gage
103	Concentricity of largest inside diameter with outside diameter		.40%	3.2	Gage
104	Second largest inside diameter, max.		.40%	3.2	Gage
105	Concentricity of second largest inside diameter with outside diameter		.40%	3.2	Gage
106	Third largest inside diameter		.40%	3.2	Gage
107	Concentricity of third largest inside diameter with outside diameter		.40%	3.2	Gage
108	Concentricity of center hole with outside diameter		.40%	3.2	Gage
109	Length from open end to largest inside basic diameter, max		.40%	3.2	Gage
110	Length from open end to second largest inside basic diameter, max.		.40%	3.2	Gage
111	Length from shoulder to open end		.40%	3.2	Gage
112	Height from shoulder to top of cap		.40%	3.2	Gage
113	Material thickness through closed end		.40%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.3	Body (prior to application of protective finish)		9215344
			NEXT HIGHER ASSEMBLY 9215343

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
114	Flatness of top of cap		.40%	3.2	Gage
115	Perpendicularity of top of cap		.40%	3.2	Gage
116	Perpendicularity of shoulder with outside dia.		.40%	3.2	Gage
117	Perpendicularity of face of open end with outside diameter		.40%	3.2	Gage
118	Length from top of cap to third largest inside basic diameter		.40%	3.2	Gage
119	Corner fillet at shoulder, max.		.40%	3.2	Gage
120	Corner radius at intersection of largest outside diameter with face of open end, max.		.40%	3.2	Gage
121	Length to larger outside basic diameter on tapered sides of cap		.40%	3.2	Gage
122	Length to smaller outside basic diameter on tapered sides of cap		.40%	3.2	Gage
123	Length from top of cap to smaller inside basic diameter, max.		.40%	3.2	Gage
124	Concentricity of tapered outer side of cap with outside diameter		.40%	3.2	Gage
125	Thickness through tapered wall of cap at indicated length from top of cap		.40%	3.2	Gage
126	Thickness through shoulder area at indicated length from open end		.40%	3.2	Gage
127	Diameter of shoulder, min.		.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.3	Body (prior to application of protective finish)		9215344
			NEXT HIGHER ASSEMBLY 9215343

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
128	Distance from top of shoulder to end of corner break, max.		.40%	3.2	Gage
129	Chamfer on inside end of center hole missing or improper		.40%	3.2	Gage
MINOR					
201	Blend radius of top of cap with tapered sides of cap improper		.65%	3.2	Visual
202	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-50546B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9215344
4.4.2.4	Body Assembly				NEXT HIGHER ASSEMBLY 9215344
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Any yellow marking present on body		100%	3.2	Visual or Gage9
2	Verification of heat treatment		100%	3.2	4.5.8
MAJOR					
101	Pull test of clinch of stud	See 4.5.7	-	3.2	4.5.7
102	Largest outside diameter, max.		.40%	3.2	Gage
103	Height of studs (2)		.40%	3.2	Gage
104	Outside diameter of stud, max. (2)		.40%	3.2	Gage
105	True position of studs		.40%	3.2	Gage
106	Diameter of center hole below noted depth		.40%	3.2	Gage
107	Length to inside end of crimp of stud, max. (2)		.40%	3.2	Gage
108	Stud, loose, missing or damaged		.40%	3.2	Visual/Manual
109	bulge or gap junction of stud with top of caps		.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-50546B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Box, Fiber, Inner (prior to insertion in bag)	1 OF 1	9206167
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined.				
MAJOR					
101	Box damaged, contents exposed		.40%	3.2	Visual
MINOR					
201	Contents loose		.65%	3.2	Manual
202	Marking misleading or unidentifiable		.65%	3.2	Visual
203	Label missing or badly wrinkled		.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-50546B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Box, Fiber, Inner (sealed)				9206166
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Barrier bag waterproofness destroyed (separation of heat seal, torn or punctured)		.40%	3.2	Visual
MINOR					
201	Marking misleading or unidentifiable		.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-50546B(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Box, Fiber, Outer (sealed)	1 OF 1	9206166
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined.				
MAJOR					
101	Box damaged, contents exposed		.40%	3.2	Visual
MINOR					
201	Contents loose		.65%	3.2	Manual
202	Marking misleading or unidentifiable		.65%	3.2	Visual
203	Label missing or badly wrinkled		.65%	3.2	Visual

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4.4.3 Testing.- Testing is described in the First Article, Quality Conformance Inspection Tables, and herein.

4.4.3.1 Profile of groove and location from open end of body (major defect).- Inspection method for this test shall be in accordance with paragraph 4.5.2. Whenever a new, repaired, or adjusted tool is entered into production the first part produced shall be tested and accepted prior to initiating production. Upon initiating production material acceptance will be based upon one of the following optional procedures.

a. Batch Processing - Parts produced on each profile tool during inspection intervals of one hour or portion thereof, shall be segregated by profile tool. After testing and acceptance of the last part produced within the interval by each tool, parts represented by the sample may be released for further processing. If any body fails to comply with any of the applicable requirements, the parts produced by the machine since the last satisfactory check shall be rejected and the tool which produced the rejected parts shall be removed from production.

b. Continuous Processing - One part shall be selected every hour or portion thereof and be subjected to this test. Failure of this part to meet the applicable requirements shall necessitate the removal of the tool which produced the part from production. All parts produced in accordance with this option shall be subjected to a subsequent inspection for presence of groove (360°).

4.4.3.2 Hardness (special defect).- Testing shall be in accordance with 4.5.3 and sampling shall be dictated by the contractor's chosen option for longitudinal and transverse strength test sampling. If the contractor samples in accordance with 4.4.3.3a he will take hardness readings for each grenade sampled from the first 500,000 grenades produced. If the contractor samples in accordance with 4.4.3.3b he will take hardness readings for each grenade sampled for the first twenty five lots of heat treatment.

4.4.3.3 Longitudinal and transverse strength (special defect).- Testing shall be in accordance with one of the following optional sampling procedures:

a. Continuous Sampling - In order to establish an acceptable heat treatment process, take a sample of fifty (50) grenade bodies for each test from each heat treat furnace at the start of production or after any interruption in the process of ninety (90) days or more. The samples shall be hardness tested in accordance with 4.5.3 (when in effect) and strength tested in accordance with 4.5.4 and 4.5.5 as applicable. Calculate the mean ($\overline{X}$) and standard deviation(s) using formula in 6.6.1. If $\overline{X}$ - 3.0S or any individual value

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falls below minimum load requirement, the process shall be corrected and an additional sample of fifty (50) grenade bodies taken. If $\bar{X}$ - 3.0S is above minimum load requirement proceed with production.

To qualify product during production a continuous sampling plan shall apply which consists of randomly selecting a body every four (4) minutes for each test from each heat treat furnace and subjecting them to the tests of paragraph 4.5.4 and 4.5.5. Bodies of the test sample of 4.5.3 may be used when that test is being conducted. The mean ($\bar{X}$) and standard deviation(s) shall be calculated baed on the value of the test sample, and individually numbered samples of forty-nine (49) grenade bodies previously tested (initial quantity obtained from above); and use of the formula in 6.6.1. The test statistics then form a running average based on fifty samples, including the current test sample as the fiftieth sample and eliminating the number one sample; (NOTE: Creates new data base in approximately three hours at full production) this process will continue for each additional body tested. If X - 3.0S limit falls below minimum load requirement or if the individual sample reading is below the minimum load specified on the drawing, reject that portion of the product since the last acceptable sample and take corrective action. A suitable hold area shall be provided at each heat treat furnace.

b. Lot Sampling - Twenty five bodies from each four hours production or portion thereof shall be randomly selected for each test. Testing shall be performed as required in 4.5.3 (when in effect), 4.5.4 and 4.5.5. Calculate the mean ($\bar{X}$) and standard deviation(S) using formula in 6.6.1. If any individual value or $\bar{X}$ - 3.0S falls below minimum value specified on the drawing the lot shall be rejected.

4.4.3.4 Phosphate coating weight (major defect).- Phosphate coating weight shall be in accordance with the applicable item drawing. Sampling and rejection shall be in accordance with TT-C-490. Testing shall be in accordance with 4.5.9.

4.4.4 Inspection equipment.- For the inspection equipment required to perform the examinations and test prescribed herein, the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, 6.3 and 6.9 herein.

4.5 Methods of inspection

4.5.1 Dimensional inspection of stock specimen.- Two (2) stock specimens each comprised of the width of the stock to be used for production, and length as obtained from the circumference of the tool to be utilized for production, shall be selected from the coil of stock prior to forming the body. One sample shall be taken from the beginning of the coil, and

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the other sample shall be taken from the end of the coil. An additional sample shall be taken whenever a new, repaired, or altered tool is entered into production. Each specimen shall be identified with an index position on the tool and for the direction of the roll. The specimen shall also be identified as being from the beginning or end of the coil. Fifteen (15) measurements shall be taken on each sample specimen for the .095 dimension and for the .112 dimension. There shall be three (3) points of measurement across the width of the specimen at five (5) places along the length. The spacing of the readings shall be approximately uniform. One (1) measurement for each characteristic shall be performed at each of the fifteen (15) measuring points on the specimen. An average value for each specimen shall be computed for each characteristic from the measurements obtained. In addition, measurements for the .1026, .1182, .010R and .025R dimensions shall be determined on the specimen from each new, repaired or adjusted rolling tool entering the production line. If the average of the fifteen (15) measurements for the .095 and .112 dimensions fails to comply, or if the direction of rolling does not comply, with the requirement specified on the applicable drawing, the coil, or portion thereof, from which the sample was taken shall be rejected and the tool from which the rejected specimen was produced shall be removed from production. If forming of the bodies has begun, those bodies represented by the rejected sample shall be rejected. All inspections shall be performed using equipment in accordance with 4.4.4.

After 8 coils have been acepted from any single embossing mill the following reduced sampling may be instituted.

At the start of production, every four (4) hours of production and at the end of production, one stock specimen shall be taken. These specimens shall come from the first coil embossed at the start of production each day and from the last coil embossed either at the end of the four hour period or at the end of production each day. The specimen shall be taken from the end of the coil and be comprised of the width of the coil and a length equal to the circumference of the embossing roller. If more than one embossing mill is used, then this procedure will be duplicated for each mill. An additional sample shall be taken whenever a new, repaired, or altered tool is entered into production. Each specimen shall be identified with an index position on the tool and for direction of roll.

Fifteen (15) measurements shall be taken on each sample specimen for the .095 dimension and for the .112 dimension. There shall be three (3) points of measurement across the width of the specimen at five (5) places along the length. The spacing of the readings shall be approximately uniform. One (1)

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measurement for each characteristic shall be performed at each of the fifteen (15) measuring points on the specimen. An average value for each specimen shall be computed for each characteristic from the measurements obtained. In addition, measurements for the .1026, .1182, .010R and .025R dimensions shall be determined on the specimen from each new, repaired, or adjusted rolling tool entering the production line. If the average of the fifteen (15) measurements for the .095 and .112 dimensions fails to comply, or if the direction of rolling does not comply with the requirement specified on the applicable drawing, that coil and the 4 hours production from which the sample was taken shall be placed in a reject status and the tool from which the rejected specimen was produced shall be removed from production, and the normal sampling will resume. Each coil produced in that four (4) hour maximum period shall be sampled in turn (end of coil only) starting with the next to the last coil produced and examined for conformance. Sampling shall terminate with the last acceptable coil. Separate four (4) hour hold areas shall be provided to satisfy inspection procedures. No coil stock shall be released for forming from the hold area before the above described inspections are satisfied. The dimensional inspections shall be performed using equipment in accordance with 4.4.4.

4.5.2 Inspection of profile of groove and location from open end of body.- The bodies to be subjected to this inspection shall be selected at the specified times from each tool used to produce the profile of the groove in the body. Each body selected shall be inspected for Depth of groove (360°). The body shall then be inspected at the locations of maximum and minimum groove depth (total of 2 places) for each of the following characteristics:

a. Length from open end of body to start of groove.
b. Length from open end of body to end of groove.
c. Depth of groove, minimum (360°).
d. Depth of groove, maximum (360°).
e. Maximum corner radius at end of groove closest to open end of body.
f. Maximum corner fillet at end of groove closest to open end of body.
g. Perpendicularity of side of groove closest to open end of body with second largest inside diameter.
h. Maximum corner radius or chamfer at intersection of largest inside diameter with face of open end of body.

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NOTE 1: Acceptance criteria for characteristics e, f and g listed above shall be as follows:

For any body subjected to this inspection, the contour of these characteristics shall lie within the zone defined by the maximum and minimum conditions of the radius, fillet and perpendicularity at the maximum and minimum conditions of groovedepth. Metal protrusion or buildup at intersection of groove corner with second largest inside diameter of body is permitted within the maximum and minimum conditions of that diameter, but this condition shall not be included in the zone defined above.

NOTE 2: The maximum depth of the groove may exceed the limit of the toleranced depth dimension up to the limit specified in the applicable drawing note providing this limit does not occur in more than one area. However, the area where the toleranced limit is exceeded shall not encompass more than one-third of the circumference of the groove.

4.5.3 Hardness test of body, prior to application of protective finish.- Each body shall be marked to identify it with the furnace from which it came. The test shall be performed using the methods and procedures specified in ASTM E-18 (part 10) and as follows: three (3) readings shall be taken on the largest outside diameter approximately 120° apart and 3/8" from the open end. These grenades shall be serialized and shall then be used for the longitudinal and transverse force tests of paragraph 4.5.4 and 4.5.5. A minimum reported data shall specify by grenade serial number the three hardness readings, their average, and corresponding results of either the transverse or longitudinal force test.

4.5.4 Longitudinal strength test of body.- The longitudinal compressive force value shall be determined in accordance with the following:

a. A universal testing machine having the capability of exerting a uniform force from 0 to 120,000 pounds shall be employed.

b. The rate of application of load shall be 0.200 plus or minus 0.025 inches per minute. This rate will be verified at the start of each production shift prior to testing of any grenades.

c. While applying the load, a continuous plot of applied force (lbs) vs. deflection (inches) shall be recorded. Recorded load range shall be 120,000 lbs. full scale and deflection magnification shall be 100X.

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d. The grenade shall be placed vertically and the load shall be applied to the shoulder of the grenade until failure occurs. The grenade shall be supported on a block of sufficient height to permit the deflectometer to rest directly on the lower platen of the testing machine with its pickup directly against the under side of the upper platen or fixture within one inch of the grenade largest outside diameter and within 1/16 inch of the centerline of the grenade. The grenade shall be placed into a fixture so that the load will be applied to the grenade shoulders only. This fixture shall have a cylindrical cut away of 1.310 ± .001 inches diameter with corners rounded to 0.0275 ± .0025 inches radius. It shall be of sufficient length to accommodate the grenade cap without interference. This fixture shall be fabricated from hardened tool steel Rockwell C 60-62. This fixture may be incorporated into the block. The upper platen or fixture shall be flat, parallel across the surfaces in contact with the part (each within 0.001 inches), and hardened to a minimum of Rockwell C 60. The block shall rest steadily on the center of the lower platen. The grenade shall be positioned such that any largest diameter will be coincident within ± 1/16 inch of the center line of the screws on the testing machine.

e. The longitudinal compressive strength shall then be determined by constructing a straight line parallel to the straight, linear portion of the trace, but offset by 0.360 inches parallel to the deflection axis of the chart. The constructed line shall then be extended until it intersects the upper nonlinear portion of the trace. The corresponding value of force thus obtained at this intersection shall represent the longitudinal yield strength of the grenade.

f. This test is a destructive test. Parts so tested shall not be returned to the production line.

4.5.5 Transverse strength test of the body.- The transverse compressive force shall be determined in accordance with the following:

a. A universal testing machine having the capability of exerting a uniform force from 0 to 20,000 pounds shall be employed.

b. The rate of application of load shall be 0.200 plus or minus 0.025 inches per minute. This rate will be verified at the start of each production shift prior to testing of any grenades.

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c. While applying the load, a continuous plot of applied force (lbs) vs. deflection (inches) shall be recorded; load ranges shall be 12,000 to 15,000 pounds full scale and deflection magnification shall be 100X.

d. The grenade shall be placed on its side and the load shall be applied along the entire length of the outside diameter until failure occurs. The grenade shall be supported on a block of sufficient height to permit the deflectometer to rest directly on the lower platen of the testing machine with its pickup directly against the under surface of the upper platen no farther than one inch from either end of the part, and on the centerline of the longitudinal axis of the part. Load applying surfaces of the block and the upper platen shall be flat, parallel across the surfaces in contact with the part (each within 0.001 inches), and hardened to a minimum of Rockwell C60. The block shall rest steadily on the center of the lower platen and its top surface shall be equipped with a nonrestraining guide for positioning the longitudinal axis of the grenade parallel to and coincident within ± 1/16 inch of the center line of the screws on the testing machine.

e. The transverse compressive strength shall then be determined by constructing a straight line parallel to the straight, linear portion of the trace, but offset by 0.30 inches parallel to the deflection axis of the chart. The constructed line shall then be extended until it intersects the upper, nonlinear portion of the trace. The corresponding value of force thus obtained or the highest value of force obtained prior to this intersection shall represent the transverse yield strength of the grenade.

f. This test is a destructive test. Parts so tested shall not be returned to the production line.

4.5.6 Inspection of body for cracks and voids.- This inspection shall be performed on each body, prior to the application of any protective coating, using the ultrasonic method. Inspection shall be performed at each of the following locations:

a. Longitudinal defects in the wall of the body from the open end.

b. Circumferential defects at the shoulder of the body originating from the interior surface.

c. Circumferential defects at the shoulder of the body originating from the exterior surface.

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d. Transverse defects throughout the entire cap wall.

e. Circumferential defects along the edge of the cap where it intersects the dome of the cap.

f. Exterior surface originating circumferential defects in the wall of the body within an area between .187 inch from the shoulder and .187 from the open end.

4.5.6.1 The following definitions shall apply for cracks and voids:

a. Crack- A discontinuity in the material with an orientation, depth and length represented by standards listed on Dwg. 9266381, and/or a discontinuity in the material which will result in an ultrasonic signal equal to or greater than that produced by the above standard after proper calibration of the ultrasonic equipment.

b. Void- A physical cavitation in the material which generates an ultrasonic signal of sufficient strength to cause rejection after calibration of equipment with standards as listed on Dwg. 9266381.

4.5.6.2 Any body which fails to comply with the requirement specified on the applicable drawing shall be classed defective and removed from the lot. Those bodies so classed as defective may be retested one additional time, using the same methods and procedures. Any body which is rejected after a retest shall be removed from the lot and destroyed.

4.5.6.3 Contractor personnel performing any calibration function of this equipment must be qualified by one of the following means:

a. Successful completion of the Level II General Examinations, ASNT-TC-1A, as given under the auspices of "The American Society for Nondestructive Testing".

b. By showing sufficient knowledge of method and familiarity with equipment to warrant approval by ARRADCOM, ATTN: DRDAR-QAR-I.

4.5.6.4 This equipment shall be verified with standards as listed on Dwg. 9226381 on a per hour frequency min. When verification indicates that the ultrasonic equipment is performing incorrectly, the quantity of bodies processed since the last satisfactory verification shall be retested.

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4.5.6.5 The calibration of this equipment shall be verified by a qualified individual prior to the start of each shift. This is to be done by sending each standard thru on a manual basis, and verifying the various electronic settings required for proper calibration.

4.5.6.6 The contractor shall make no changes in equipment or procedures without prior written approval of the Procuring Contractor Officer.

4.5.7 Stud loose and pull test of clinch of stud on body assembly.- As a minimum one body assembly shall be sampled from each machine from every 3800 bodies assembled starting with the first unit produced at each production shift and whenever a new, repaired or adjusted crimping tool is put into use for production. Each sampled body assembly shall be inspected manually first for loose studs and gap at junction of studs with top of cap. Any body assembly passing this test shall be placed in an approved fixture which will restrain the body. A tensile force, in an axial direction, shall then be applied to one of the studs and the stud must withstand the minimum force specified on the drawing. The test shall be repeated using the other stud on the body assembly. If any body assembly fails to comply with either requirement (loose stud or pull test), production from the tool which yielded the defective part since the last satisfactory test, shall be rejected. Bodies from this test may be reworked to assemble new studs provided that the two stud holes are within the dimensional requirements of the drawing after the removal of the original studs and that the protective finish on the body has not been affected to expose the base metal during the rework process. Adequate hold areas shall be provided at each machine to accommodate parts produced between inspections. Accepted parts shall be released into the line for other inspections on the body assembly.

4.5.8 Verification of heat treatment of body assembly.- This verification test shall be performed on each body assembly immediately prior to packaging. This inspection shall employ the eddy current method using a circular coil with its longitudinal axis parallel to the longitudinal axis of the grenade and the circular cross section of the grenade centered within the coil. However, any other equivalent method approved by the procuring agency and DRDAR-QAR-I may be used. (Advisory: Acceptable parts shall have a hardness which falls between R_A68 and RA73). If any component fails to comply with the requirement specified on the applicable drawing, it shall be classified defective and removed from production.

4.5.9 Phosphate coating weight.- The test method shall be in accordance with TT-C-490.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.- The body assembly shall be preserved and packaged in accordance with Dwg. 9206167.

5.2 Packing

5.2.1 Level C.- The body assembly, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with Dwg. 9206166.

5.3 Marking.- Marking shall be in accordance with drawings 9206167 and 9206166.

5.4 Shipping.- When components from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use.- The body assembly is intended for use in the M42 Grenade.

6.2 Ordering data.- See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval.- See MIL-A-48078. Submit equipment designs as required to: Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Contractor's written control procedure for the overall heat treatment process.- The contractor shall prepare a written plan to be implemented for control of his overall heat treatment process to facilitate homogeneity of the heat treated bodies. The written plan shall specify, as a minimum, controls and procedures for the following:

a. Heat treatment temperature tolerance.- The maximum temperature deviation to be permitted from the pre-determined temperature for all phases in the heat treatment process including the permitted temperature range of the quench medium before and during the quench operation.

b. Preheat.- The temperature and method of preheat, including designation of the phases(s) to which the method shall apply.

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c. Confirmation of accuracy of temperature-measuring instruments.- The method, location and frequency basis for obtaining temperature checks for all the temperature-recording instruments of all the heat treatment phases.

d. Check for detrimental surface effects due to salt or gaseous contamination.- The test method, criteria, frequency basis and heat treatment phase source for determining the presence of detrimental surface effects.

e. Elapsed time between hardening and quench phases.- The maximum time interval to be permitted and the method of checkin, from the end of the hardening phase to complete submersion in the quench medium.

f. Elapsed time between quench and tempering phases.- The maximum time interval to be permitted between these phases and the medium of checking, including temperature of the bodies, prior to tempering.

g. Nested bodies.- The policy and handling methods to be employed with regard to nested bodies, particularly for the quench phase and the method of checking for compliance.

h. Prevention of mixing non or partially heat treated bodies with those fully heat treated.- Procedures shall specify steps taken to preclude such mixing of bodies caused by sticking in handling containers, in-process inspections, poor housekeeping or any other condition by which mixing of bodies may occur.

i. Records.- The storage of permanent records, available for government review of the time temperature cycle for all phases of heat treatment process and for the items in (b) through (g), above. The procedure shall provide for the identification of such records by heat treatment batch and lot number, as applicable.

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6.5 Data requirements.- Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
6.4	Procedure Report	*DI-T-5372 (Tailored)
4.5.2, 4.5.4, 4.5.7	Quality Inspection Test, Demonstration and Evaluation Rpt	DI-R-1721

* Delete a, b, c, d, of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits.

The above data shall be forwarded in accordance with the requirements of the contract to Commander, ARRADCOM, ATTN: DRDAR-QAR-Q, Dover, NJ 07801

6.6 Strength test of the body

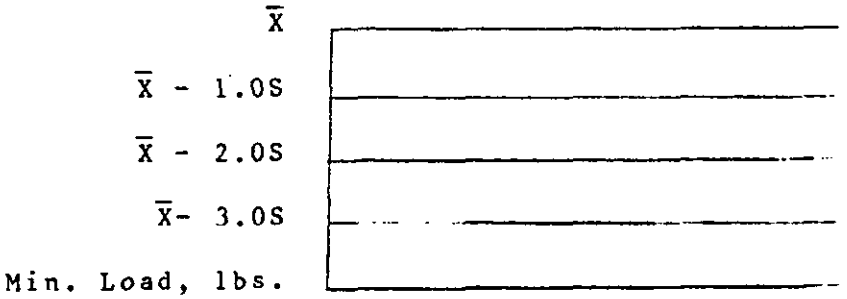
6.6.1 Calculation of sample standard deviation(s).- The following formula or equivalent shall be used in calculating the sample standard deviation:

$$S = \sqrt{\frac{\Sigma (\overline{X} - X)^2}{N - 1}}$$

Where: S = Sample standard deviation.
X = Value of each individual observation, lbs.
$\overline{X}$ = Average value of all individual observation, lbs.
N = Sample size.

6.6.2 Control chart.- If $\overline{X}$ - 3.0S limit (ref. 4.4.3.2a) is above minimum load requirement setup a control chart as shown below:

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6.6.3 Recommended procedure to monitor the process.- The contractor shall consider the process out of control if any of the following occur:

a. Any point falls outside the $\overline{X}$ - 3.0S limit.

b. If a run of seven (7) consecutive points fall below the mean.

c. If a run of seven (7) consecutive points are decreasing.

d. A run of two (2) consecutive points below the $\overline{X}$ - 2.0S limit.

e. A run of four (4) consecutive points below the $\overline{X}$ - 1.0S limit.

f. If any of the above (a thru e) occurs corrective action shall be taken, and an acceptable heat treatment process established in accordance with paragraph 4.4.3.3.a.

6.6.4 Hardness test of body.- The test is utilized to provide data for determining a possible correlation of hardness with longitudinal and transverse strength values.

6.7 Drawings.- Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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6.8 Submission of test data.- In addition to the normal distribution of records, when the body assembly is procured by the US Army Armament Materiel Readiness Command, one (1) copy of all Ammunition Data Cards shall be forwarded to: Commander, US Army Armament R&D Command, ATTN: DRDAR-QAR-Q, Dover, New Jersey 07801.

6.9 Automatic inspection equipment.- All automatic test and inspection equipment when utilized by the contractor shall be subjected to a periodic verification and/or a calibration check at frequent intervals during normal production. The procedures shall include as a minimum; frequency, method of verification, and the method of retrieval of item produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by Paragraph 4.4.4.

6.11 Alternate inspection procedures.-

6.11.1 Justification for alternate inspection procedures.- Submitted documentation must include sufficient information to indicate an economic benefit, and must also clearly identify that there is no reduction in product quality.

6.11.2 Alternate inspection procedures plan.- The alternate inspection procedure plan as a minimum must include:

a. Frequency (production interval) for validating equipment functioning.

b. Proposed method of validation to include process and inspection drawings.

c. Description of material control procedures to be imposed during the production interval between validations (Note: Under these provisions final acceptance will be withheld from material produced during a production interval until conducting satisfactory validation.)

6.11.3 Approval of alterntate inspection procedures.- Proposed plans shall be submitted per para. 6.5 for approval.

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-G-50546B GRENADE, GP: M42 BODY ASSEMBLY, TYPE PH, FOR

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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